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ASSOCIATE DIRECTOR, CENTER FOR GENOMIC MEDICINE
PHYSICIAN, PULMONARY & CRITICAL CARE UNIT
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Education & Training

	Whitehead Institute for Biomedical Research
2018 - 2022	Postdoctoral research fellow (advisor: Prof. David M. Sabatini)
	Massachusetts General Hospital & Harvard Medical School
2016 - 2020	Clinical fellow, Pulmonary and Critical Care Medicine
2014 - 2016	Resident physician, Internal Medicine
2013 - 2014	Resident physician, General Surgery
	Johns Hopkins University
2007 - 2013	Doctor of Philosophy in Human Genetics (advisor: Prof. Joshua T. Mendell)
2005 - 2013	Doctor of Medicine Alpha Omega Alpha; Medical Scientist Training Program
2001 - 2005	Bachelor of Arts in Neuroscience Phi Beta Kappa; University and Departmental Honors

Honors & Awards

2026	Howard M. Goodman Fellowship	Mass General Brigham / Department of Molecular Biology
2026	BWH-MGB Collaborative IGNITE Award	Brigham and Women's Hospital / Massachusetts General Hospital
2025	Mark C. Fishman, M.D. Teaching Award	Massachusetts General Hospital
2024	Dept. of Medicine Transformative Scholar Award	Massachusetts General Hospital / Tianqiao and Chrissy Chen Institute
2024	Award for Excellence in Biomedical Research	Smith Family Foundation
2023	Ellison Foundation Award	Ellison Foundation
2023	Young Physician-Scientist Award	American Society of Clinical Investigation
2022	Donahoe Catalyst Award	Massachusetts General Hospital Department of Surgery
2021	Career Award for Medical Scientists (CAMS)	Burroughs Wellcome Fund
2021	Basic Science Prize	Massachusetts Pulmonary Society
2020 - 2023	Parker B. Francis Fellowship Award	Parker B. Francis Family Foundation
2019	ATS Foundation Unrestricted Grant Award	American Thoracic Society Foundation
2019	First place award in basic science research	National Jewish Respiratory Disease Young Investigators Forum
2019	Peer award for best overall presentation	National Jewish Respiratory Disease Young Investigators Forum
2019	Abstract Scholarship	Respiratory Cell/Molecular Biology Section, American Thoracic Society
2019	Winner, Koch Institute Image Awards	Massachusetts Institute of Technology
2018	2nd prize, W.M. Keck Image Awards	Whitehead Institute for Biomedical Research
2018 - 2020	NIH Kirschstein Institutional NRSA Award	Massachusetts General Hospital
2015	Massachusetts champion, "Doctor's Dilemma"	American College of Physicians
2014	"30 Under 30" in Science and Healthcare	Forbes Magazine
2013	Michael A. Shanoff Research Award	Johns Hopkins University School of Medicine
2013	Alpha Omega Alpha	Johns Hopkins University School of Medicine
2013	William Stewart Halsted Award in Surgery	Johns Hopkins University School of Medicine
2009	2nd prize, Best Ph.D. Student Talk	IMBA Small RNA Microsymposium (Vienna, Austria)
2006 - 2013	NIH Medical Scientist Training Program Award	National Institutes of Health
2005	Dean's Summer Research Award	Johns Hopkins University School of Medicine
2004	HHMI undergraduate research fellowship	Johns Hopkins University
2004	Phi Beta Kappa	Johns Hopkins University
2004	Inter-fraternity Council Outstanding Junior	Johns Hopkins University
2003	Mensa Member Scholarship	Mensa Education & Research Foundation
2003	Provost's Undergraduate Research Award	Johns Hopkins University
2000	National Merit Scholarship Finalist	Wichita High School East (Wichita, KS)

Scientific contributions

[papers]

[2020 -]

Valenstein ML*, Wranik M*, Lalgudi PV, Linde-Garelli KY, Choi Y, **Chivukula RR**, Sabatini DM, Rogala KB. Structural basis for the dynamic regulation of mTORC1 by amino acids. *Nature*. doi: 10.1038/s41586-025-09428-7. (2025)

Ziehr DR, **Chivukula RR**. From Protein Folding to Precision Therapy: SP-A in the Spotlight. *Am J Respir Cell Mol Biol*. doi: 10.1165/rcmb.2025-0366ED. (2025)

Yang J, Xu Y, Ziehr DR, Taylor MS, Valenstein ML, Frenkel EM, Bush JR, Rutter K, Stevanovski I, Shi CY, Kesavan M, Mouro Pinto R, Deveson I, Bartel DP, Sabatini DM, **Chivukula RR**. Polyglycine-mediated aggregation of FAM98B disrupts tRNA processing in GGC repeat disorders. *Science*. ado2403 (2025). doi: 10.1126/science.ado2403.

Valenstein ML, Lalgudi PV, Kedir JF, Condon KJ, Platzek A, Freund DG, Taylor MS, Xu Y, **Chivukula RR**, Sabatini DM. Amino acids and KLHL22 do not activate mTORC1 via DEPDC5 degradation. *Nature*. Jan;637(8045) (2025)

McGarrity S., Ziehr D.R., Austin-Tse C.A., Wein M.N., **Chivukula R.R.**, Oldham W.M. Exercise Intolerance and Low Cardiac Filling Pressures in a Woman With a Novel eNOS Mutation. *Circ Genom Precis Med*. 17(5):e004741. doi: 10.1161/CIRCGEN.124.004741 (2024)

Valenstein ML*, Lalgudi PV*, Gu X*, Kedir JF, Taylor MS, Xu Y, **Chivukula RR**, Sabatini DM. Rag-Ragulator is the central organizer of the physical architecture of the mTORC1 nutrient-sensing pathway. *Proc Natl Acad Sci*. 121(35):e2322755121. doi: 10.1073/pnas.2322755121 (2024)

Shi C.Y., Elcavage L.E., **Chivukula R.R.**, Stefano J., Kleaveland B., Bartel D.P. ZSWIM8 destabilizes many murine microRNAs and is required for proper embryonic growth and development. *Genome Research*. 33(9):1482-1496. doi: 10.1101/gr.278073.123. (2023)

Boehnke N., Straehla J.P., Safford H.C., Kocak M., Rees M.G., Ronan M., Adelmann C.H., **Chivukula R.R.**, Cheah J.H., Li, H., Sabatini D.M., Roth J.A., Koehler A.N., Hammond P.T. Massively parallel pooled screening reveals genomic determinants of nanoparticle-cell interactions. *Science*. eabm5551 (2022). doi: 10.1126/science.abm5551

Eshel N. & **Chivukula RR.** Rethinking the Physician-Scientist Pathway. *Academic Medicine*. 97(9):1277-1280. doi: 10.1097/ACM.0000000000004788 (2022)

Shah V.S., **Chivukula R.R.**, Lin B., Waghray A., Rajagopal J. Cystic fibrosis and cells of the airway epithelium: What are ionocytes and what do they do? *Annual Review of Pathology: Mechanisms of Disease*. doi: 10.1146/annurev-pathol-042420-094031 (2022)

Chivukula R.R., Maley J.H., Dudzinski D.M., Hibbert, K.A., Hardin, C.C. Evidence-based management of the critically ill adult with SARS-CoV-2 infection. *Journal of Intensive Care Medicine*. 36(1):18-41 (2021).

Hariri, L.P., North, C.M., Shih, A., Villalba, J., Vinarsky, V., Hardin, C.C., Rubin, J., Okin, D.A., Sclafani, A., Alladina, J.W., Griffith, J.W., Gillette, M.A., Raz, Y., Richards, C.J., Wong, A.K., Ly, A., Hung, Y.P., **Chivukula, R.R.**, [...] Stone, J.R., Mino-Kenudson, M. Lung histopathology in COVID-19 as compared to SARS and H1N1 influenza: a systematic review. *CHEST*. 159(1):73-84 (2020).

Maley, J.H., Petri, C.R., Brenner, L.N., **Chivukula, R.R.**, […] Hardin, C.C. Anticoagulation, Immortality, and Observations of COVID-19. *Research and Practice in Thrombosis and Haemostasis*. 4 (5), 674-676 (2020).

Petri C.R., Brenner L.N., Calhoun T.F., **Chivukula R.R.**, Maley, J.H. Vinarsky V., Dudzinski D.M., Hardin C.C. The FLARE Initiative: A collaborative effort for timely literature appraisal during the COVID-19 pandemic. *ATS Scholar*. doi: 10.34197/ats-scholar.2020-0045IN (2020)

Shamseldin, H.E., Mogarri, I., Alqwaiee, M., Alharbi, A., Baqais, K., AlSaadi, M., AlAnzi, T., Alhashem, A., Saghier, A., Ameen, W., Ibrahim, N., Yang, J., Abdulwahab, F., Hashem, M., **Chivukula, R.R.**, Alkuraya, F.S. An exome-first approach to aid in the diagnosis of primary ciliary dyskinesia. *Human Genetics*. 139(10):1273-1283 (2020).

Chivukula, R.R.†, […] Alkuraya, F.S.†, Sabatini, D.M. A human ciliopathy reveals essential functions for NEK10 in airway mucociliary clearance. *Nature Medicine*. 26(2), 244-251 (2020). †*corresponding*

[2010 - 2020]

Taylor, M.S.*, **Chivukula, R.R.***, […] Kradin R.L. A conserved distal lung regenerative pathway in acute lung injury. *Am J Pathol*. 188(5):1149-1160 (2018). **equal contribution*

Taylor, M.S.*, **Chivukula, R.R.***, […] Kradin R.L. Delayed alveolar epithelialization: a distinct pathology in diffuse acute lung injury. *Am. J. Respir. Crit. Care Med*. 197, 522–524 (2018). **equal contribution*

Knabel, M.K., Ramachandran, K., Karhardkar, S., Hwang, H.W., Creamer, T.J., **Chivukula, R.R.**, […] Warren D.S. Systemic delivery of scAAV8-encoded miR-29a ameliorates hepatic fibrosis in carbon tetrachloride-treated mice. *PLoS ONE* 10, e0124411 (2015).

Zeitels, L.R., Acharya, A., Shi, G., Chivukula, D., **Chivukula, R.R.**, […] Mendell, J.T. Tumor suppression by miR-26 overrides potential oncogenic activity in intestinal tumorigenesis. *Genes Dev*. 28, 2585-2590 (2014).

Chivukula, R.R.*, Shi G.*, […] Mendell, J.T. An essential mesenchymal function for miR-143/145 in intestinal epithelial regeneration. *Cell* 157, 1104-16 (2014). **equal contribution*

Hsu S., Wang B., Kota J., Yu J., Costinean S., Kutay H., Yu L., Bai S., LaPerle K., **Chivukula R.R.**, […] Ghoshal K. Essential metabolic, anti-inflammatory and anti-tumorigenic functions for miR-122 in mouse liver. *J. Clin. Invest*. 122, 2871-2883 (2012).

Pramanik, D., Campbell N., Karikari C., **Chivukula R.**, […] Maitra A. Restitution of tumor suppressor microRNAs using a systemic nanovector inhibits pancreatic cancer growth in mice. *Mol. Cancer Ther*. 10, 1470–1480 (2011).

Kent, O.A., **Chivukula R.R.**, […] Mendell J.T. Repression of the miR-143/145 cluster by oncogenic Ras initiates a tumor-promoting feed-forward pathway. *Genes Dev*. 24, 2754–2759 (2010).

Persat, A., **Chivukula, R.R.**, Mendell, J.T. & Santiago, J.G. Quantification of global microRNA abundance by selective isotachophoresis. *Anal. Chem*. 82, 9631–9635 (2010).

[2000 - 2010]

Kota, J., **Chivukula R.R.**, […] Mendell J.T. Therapeutic microRNA delivery suppresses tumorigenesis in a murine liver cancer model. *Cell* 137, 1005–1017 (2009).

Chivukula, R.R. & Mendell, J.T. Abate and switch: miR-145 in stem cell differentiation. *Cell* 137, 606–608 (2009).

Soderlund, K.A., **Chivukula R.R.**, […] Halushka M.K. Prognostic value of left ventricular apical tissue removed for HeartMate II left ventricular assist device placement. *Cardiovasc. Pathol*. 18, 217–222 (2009).

Chang, T.-C., Zeitels L.R., Hwang H., **Chivukula R.R.**, […] Mendell J.T. Lin-28B transactivation is necessary for Myc-mediated let-7 repression and proliferation. *Proc. Natl. Acad. Sci. USA* 106, 3384–3389 (2009).

Allen, G., Simpkins C.E., Segev D., Warren D., King K., Taube J., Locke J., Baldwin W., Haas M., **Chivukula R.** & Montgomery R.A. Rapid accommodation of an A1 renal allograft after preconditioning for ABO-incompatible transplantation. *Contrib. Nephrol.* 162, 35–46 (2009).

Chivukula, R.R. & Mendell, J.T. Circular reasoning: microRNAs and cell-cycle control. *Trends Biochem. Sci.* 33, 474–481 (2008).

Chaudhury, S., Eisinger J., Hao L., Hicks J. **Chivukula R.** & Turano K. Visual illusion in virtual world alters women's target-directed walking. *Exp. Brain. Res.* 159, 360–369 (2004).

[patents]

Chivukula, R.R. & Sabatini D.M. "The kinase NEK10 and its use in treating and diagnosing bronchiectasis and other respiratory disorders". U.S. Patent WO2021146704.

Mendell, J.T., Mendell, K.A., **Chivukula, R.R.**, et. al. "Compositions and methods for treating hepatic neoplasia." U.S. Patent 8,729,041. 20 May 2014.

[books/chapters]

Chang MG, Luckhurst CM, **Chivukula RR**, Dudzinski DM, Izzy S, Bittner EA, eds. Massachusetts General Hospital Review of Critical Care Medicine. 2nd ed. Philadelphia: Wolters Kluwer; 2026.

Chivukula RR. Pulmonary mechanics. In: Chang MG, Luckhurst CM, Chivukula RR, Dudzinski DM, Izzy S, Bittner EA, eds. Massachusetts General Hospital Review of Critical Care Medicine. 2nd ed. Philadelphia: Wolters Kluwer; 2027:91.

Chivukula RR. Respiratory failure, mechanical ventilation, and weaning. In: Chang MG, Luckhurst CM, Chivukula RR, Dudzinski DM, Izzy S, Bittner EA, eds. Massachusetts General Hospital Review of Critical Care Medicine. 2nd ed. Philadelphia: Wolters Kluwer; 2027:92.

Chivukula RR. Pleural disease. In: Chang MG, Luckhurst CM, Chivukula RR, Dudzinski DM, Izzy S, Bittner EA, eds. Massachusetts General Hospital Review of Critical Care Medicine. 2nd ed. Philadelphia: Wolters Kluwer; 2027:121.

Chivukula RR. Pulmonary vasculitic and autoinflammatory disorders. In: Chang MG, Luckhurst CM, Chivukula RR, Dudzinski DM, Izzy S, Bittner EA, eds. Massachusetts General Hospital Review of Critical Care Medicine. 2nd ed. Philadelphia: Wolters Kluwer; 2027:129.

Chivukula RR. Inhalational injuries and tracheobronchitis in the intensive care unit. In: Chang MG, Luckhurst CM, Chivukula RR, Dudzinski DM, Izzy S, Bittner EA, eds. Massachusetts General Hospital Review of Critical Care Medicine. 2nd ed. Philadelphia: Wolters Kluwer; 2027:133.

Chivukula RR. Sleep-related breathing disorders. In: Chang MG, Luckhurst CM, Chivukula RR, Dudzinski DM, Izzy S, Bittner EA, eds. Massachusetts General Hospital Review of Critical Care Medicine. 2nd ed. Philadelphia: Wolters Kluwer; 2027:139.

Chivukula RR. Hematology diagnostic modalities. In: Chang MG, Luckhurst CM, Chivukula RR, Dudzinski DM, Izzy S, Bittner EA, eds. Massachusetts General Hospital Review of Critical Care Medicine. 2nd ed. Philadelphia: Wolters Kluwer; 2027:257.

Magder, S. and **Chivukula, R.**, (2021) 'Acid Base and Hydrogen Ion', in Magder S., Malhotra A., Hibbert K.,, Hardin C., (eds) Cardiopulmonary Monitoring: Basic Physiology, Tools, and Bedside Management for the Critically Ill. Switzerland: Springer.

Le T., Bechis S., eds. **Chivukula R.**, et. al. (2009) *First Aid Q&A for the USMLE Step I*. New York: McGraw-Hill/Medical.

[selected talks]

"Translation Stress and Proteostasis Defects: Insights from Rare Disease" *Harvard Combined PCCM Fellowship Grand Rounds* (2025)

"Translation Stress and Proteostasis Defects: Insights from Rare Disease Genetics" *Tianqiao and Chrissy Chen Institute Symposium* (2025)

"Causes and consequences of tRNA splicing disorders in neurological disease" *Kansas State University Department of Biochemistry and Biophysics* (2025)

"Genetic disorders of proteostasis: Pathogenic mechanisms in GGC repeat expansions" *Mass General Brigham Department of Medicine Research Seminar Series* (2025)

"Leveraging rare respiratory disorders to illuminate pulmonary pathophysiology" *Massachusetts Pulmonary Society Conference* (2023)

"Leveraging rare respiratory disorders to illuminate pulmonary physiology" *Pulmonary Basic & Translational Research Conference, Weill Cornell Medical Center* (2023)

"New roles for polyglycine in health and disease" *Center for Genomic Medicine seminar series, Massachusetts General Hospital* (2022)

"Leveraging rare respiratory disorders to illuminate pulmonary physiology" *FASEB Lung Epithelium Conference* (2022)

"Leveraging rare respiratory disorders to illuminate pulmonary physiology" *Faculty Search Seminar, Center for Genomic Medicine, Massachusetts General Hospital* (2022)

"Leveraging rare respiratory disorders to illuminate pulmonary physiology" *Institute Seminar, Cardiovascular Research Institute (CVRI) / Lung Biology Center, University of California San Francisco (UCSF) Medical Center* (2021)

"Leveraging rare respiratory disorders to illuminate pulmonary physiology" *Division of Pulmonary Medicine, Boston Children's Hospital* (2021)

"Leveraging rare respiratory disorders to illuminate pulmonary physiology" *Children's Research Institute lecture, University of Texas Southwestern Medical Center* (2021)

"Leveraging rare respiratory disorders to illuminate pulmonary physiology" *University of Pennsylvania / Children's Hospital of Philadelphia (Penn-CHOP) Lung Biology Seminar* (2021)

"Leveraging rare respiratory disorders to illuminate pulmonary physiology" *Visiting Professor Grand Rounds, Division of Allergy, Pulmonary, & Critical Care Medicine, Vanderbilt University Medical Center* (2021)

"A human ciliopathy reveals essential functions for NEK10 in airway mucociliary clearance" *MA Pulmonary Section, American Lung Association* (2021)

"New tools for probing lysosome function and dysfunction *in vivo*" *Hermansky-Pudlak Syndrome (HPS) Network Conference* (2021)

"A human ciliopathy reveals essential functions for NEK10 in airway mucociliary clearance" *Department of Medicine Grand Rounds, Massachusetts General Hospital* (2020)

"Leveraging rare respiratory disorders to illuminate pulmonary physiology" *Children's Research Institute lecture, University of Texas Southwestern Medical Center* (2020)

'NEK10 controls human airway mucociliary clearance." *University of Iowa Airway Research Seminar* (2019)

"Inactivating mutations in NEK10 cause a human short cilia syndrome." *National Jewish Respiratory Disease Young Investigators' Forum* (2019)
"NEK10 controls human airway mucociliary clearance." *Whitehead Institute for Biomedical Research Forum seminar* (2019)
"NEK10 loss-of-function causes a novel human motile ciliopathy." *Rare Lung Diseases mini-symposium, American Thoracic Society Conference* (2019)
"Molecular genetics of primary ciliary dyskinesias." *Grand Rounds, Division of PCCM, MGH.* (2017)
"The science of the night: novel and unexpected roles for miR-143/145 in intestinal regeneration." *Department of Medicine Grand Rounds, Massachusetts General Hospital* (2016)
"Novel and unexpected roles for miR-143/145 in mouse intestinal regeneration." *The Johns Hopkins University School of Medicine Young Investigator's Day Michael Shanoff Award Lecture.* (2013)
"Therapeutic microRNA replacement in murine liver cancer." *microRNAs in Human Disease Meeting.* (2010)
"Therapeutic microRNA replacement in murine liver cancer." *IMBA Microsymposium on Small RNAs.* (2009)

Other experience & certification

[experience] Ad hoc referee: *Science*, *Nature*, *Journal of Intensive Care Medicine*, *Acta Anaesthesiologica Scandinavica, American Journal of Physiology- Lung Cellular and Molecular Physiology, Genetics in Medicine, American Journal of Respiratory Cell & Molecular Biology*

Co-Director, Adult Pulmonary Genetics Clinic, Massachusetts General Hospital (2023-)
Pathways Consult Service Core Scientist, Massachusetts General Hospital (2021-)
Core Faculty, BBS 230 Course, Harvard Medical School (2025-)
Co-Director, Primer on Complex Trait Genetics, Mass General Hospital and Broad Institute of Harvard and MIT (2023, 2025)
Lecturer, Human Physiology Course, Massachusetts Institute of Technology (2022-)
House staff critical care lecture faculty, Massachusetts General Hospital (2019-)
Faculty, Harvard "Principles of Critical Care Medicine for Non-Intensive Care Specialists" (2017-2019)

Associate Director, Center for Genomic Medicine, Massachusetts General Hospital (2026-)
Infrastructure Committee, Center for Genomic Medicine, Massachusetts General Hospital (2023-)
Seminar Committee, Center for Genomic Medicine, Massachusetts General Hospital (2023-2025)
Training grant review committee, Division of Pulmonary and Critical Care Medicine, Massachusetts General Hospital (2020-)

[certification] Diplomate in Critical Care Medicine, American Board of Internal Medicine (2021)
Diplomate in Pulmonary Disease, American Board of Internal Medicine (2020)
Diplomate in Internal Medicine, American Board of Internal Medicine (2017)
Massachusetts state medical license, MA Board of Registration in Medicine (2017)